Azure & Agent Foundations

From cloud basics to your first team of AI agents in Microsoft 365 Copilot

Azure foundations | How AI agents work | Agent Builder, Notebooks, Cowork, Skills | Chaining agents safely

Simple concepts, no coding needed

Hands-on labs every day

A human approves every decision

CodeSizzler · Microsoft Preferred Partner, Data & AI
www.codesizzler.in

Meet Your Instructor

A practitioner who designs, builds and runs Microsoft Data & AI solutions for enterprises in the region.



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Ask questions any time. This week is a conversation, not a lecture.

01

Microsoft MVP (Azure)

Recognised by Microsoft for community leadership and deep Azure expertise.

02

Microsoft Certified Trainer

Certified to deliver official Microsoft training and labs.

03

What I build

AI agents, Microsoft Fabric, Copilot Studio, Power Platform, cloud migration, enterprise data architecture.

04

Where I work

Clients across Saudi Arabia, the UAE, Singapore, the US and the UK.

Five Days from Cloud Basics to Production-Ready AI

Each day adds one layer. Every day mixes short concepts, a live demo and a hands-on lab.

DAY 1 · TODAY

Azure & Agent Foundations

Agentic workflows with Microsoft 365 Copilot

- Azure cloud foundations
- How AI agents work
- Agent Builder, Notebooks, Cowork, Skills
- Chaining agents with a human in the loop

DAY 2

Copilot Studio

Low-code and pro-code conversational AI

- Agents, workflows and apps
- Knowledge, MCP tools, skills
- Topics and the standard harness
- Evaluate, publish, govern

DAY 3

Microsoft Foundry

Azure AI foundation and core concepts

- Projects, resources, endpoints
- Model catalog and comparison
- Prompting and evaluation
- Secure, private, observable agents

DAY 4

Data, RAG & Power BI

Grounding AI in trusted data

- Data quality for AI
- Retrieval-augmented generation
- Azure AI Search patterns
- Power BI basics and dashboards

DAY 5

Microsoft Fabric

Enterprise analytics with Copilot

- OneLake, lakehouse, shortcuts
- Copilot in Fabric and Power BI
- Fabric IQ ontology and data agents
- Governance and the road ahead

Foundations → Microsoft 365 Copilot → Copilot Studio → Foundry → RAG & Power BI → Fabric

How the Week Fits Together

Four Microsoft AI platforms, one data foundation, all running on Azure.

Microsoft 365 Copilot	Personal and team assistants over your mail, files, Teams and SharePoint (Day 1)
Copilot Studio	Low-code agents, workflows and apps for departments and the company (Day 2)
Microsoft Foundry	Pro-code agents, models, evaluations and private networking (Day 3)
RAG & Power BI	Grounding answers in documents; dashboards on clean data (Day 4)
Microsoft Fabric	OneLake, Copilot in Fabric, Fabric data agents over plant data (Day 5)
Microsoft Azure	Identity, security, networking, compute and storage under everything (Day 1)

What you leave with on Friday	
Day 1	Two chained agents you built yourself
Day 2	A service-desk agent with MCP tools
Day 3	A Foundry project with evaluated models
Day 4	A clean dataset and a Power BI dashboard
Day 5	A refinery ontology and data agent
Always	A clear rule for which platform to use

Today: Three Parts, One Goal

Understand the foundation, understand agents, then build and chain your own.

01 Azure Foundations → 02 Agent Foundations → 03 M365 Copilot Agents → 04 Hands-on Labs

Part 1 · Concepts

- Cloud basics and service models
- Regions, subscriptions, resource groups
- Identity, security, networking
- Compute, data, AI services, cost

Part 2 · Concepts

- From LLM to agent
- Knowledge, tools, memory, orchestration
- MCP, multi-agent patterns
- Human in the loop and guardrails

Part 3 · Demo

- Copilot Chat, Notebooks, Cowork, Skills
- Agent Builder walkthrough
- Writing instructions that refuse to guess

Part 4 · Labs

- Lab 1: Launch Coordination chain
- Lab 2: Governance layer
- Challenge: Customer Care Agent Chain

Before we start

- Microsoft 365 account with Copilot
- Edge or Chrome
- Access to OneDrive
- Lab files from your session lead
- Permission to create agents

By the end of today, every participant has built and chained two agents.

01

Microsoft Azure Foundations

The cloud platform under every AI service we use this week

- Cloud concepts
- Global infrastructure
- Identity & security
- Core services
- Cost & good practice

What Is Cloud Computing?

Renting computing power over the internet, the way you buy electricity from the grid.

Cloud computing delivers servers, storage, databases, networking, software and AI over the internet, on demand.

- **On demand:** create a server or database in minutes instead of waiting weeks for hardware.
- **Pay for what you use:** like an electricity bill, you are charged for consumption, not for owning the power station.
- **Scale up and down:** add capacity for a month-end peak, then release it.
- **Global reach:** run close to your users and your data residency requirements.
- **Managed for you:** the provider runs the data centres, hardware and much of the software.

THREE DEPLOYMENT MODELS

- **Public cloud:** shared Microsoft data centres, e.g. Azure
- **Private cloud:** dedicated to one organisation
- **Hybrid:** both, connected together; common in refineries with on-site plant systems

Most enterprises, including industrial ones, run hybrid.

Why Organisations Move to the Cloud

Six benefits you will hear in every cloud conversation.

01 High availability

Services keep running even when a component fails, backed by service-level agreements.

02 Scalability

Add more resources (scale out) or bigger ones (scale up) when demand grows.

03 Elasticity

Scale automatically with demand, then shrink back so you stop paying.

04 Reliability

Data and services can be replicated across zones and regions to survive outages.

05 Security

Built-in identity, encryption and threat protection, operated by specialist teams.

06 Predictability

Forecast performance and cost with tools like the pricing calculator and budgets.

The cloud does not remove responsibility. It changes who does what.

CapEx vs OpEx: How You Pay Changes

The consumption model turns large up-front purchases into a monthly operating cost.

Topic	On-premises (CapEx)	Cloud (OpEx)
Up-front cost	✗ Buy servers, storage and licences before you start	✓ No up-front hardware; start small and pay monthly
Capacity planning	✗ Size for the peak and hope you guessed right	✓ Match capacity to real demand, minute by minute
Time to start	✗ Weeks to months for procurement and setup	✓ Minutes to create a resource
Idle capacity	✗ You pay for it whether it is used or not	✓ Stop or delete it and the meter stops
Experiments	✗ Expensive to try and expensive to fail	✓ Cheap pilots: try an AI idea for a few dollars

For AI this matters: you can pilot an agent cheaply before committing to scale.

IaaS, PaaS and SaaS in Plain Words

The higher you go, the less you manage yourself.

IaaS · Infrastructure

You rent the building blocks.

- Virtual machines, disks, networks
- You manage the OS, patches and apps
- Most control, most effort
- Azure example: Azure Virtual Machines

PaaS · Platform

You bring the app; Azure runs the platform.

- No servers to patch
- Focus on code, data and configuration
- Balanced control and effort
- Azure examples: App Service, Azure SQL Database, Azure AI Search

SaaS · Software

You just use the finished service.

- Sign in and work
- Microsoft runs everything underneath
- Least effort, least control
- Examples: Microsoft 365 Copilot, Power BI, Fabric

This week: Microsoft 365 Copilot and Fabric are SaaS; Foundry and AI Search are PaaS.

Who Is Responsible for What?

Security in the cloud is shared. Some duties never move away from you.

Responsibility	On-premises	IaaS	PaaS	SaaS
Data, devices, accounts and identities	You	You	You	You
Applications	You	You	Shared	Microsoft
Network controls	You	You	Shared	Microsoft
Operating system	You	You	Microsoft	Microsoft
Physical hosts, network and data centre	You	Microsoft	Microsoft	Microsoft

Your data, your users and who can access what always remain your responsibility.

Regions, Availability Zones and Data Residency

Where your data and AI models physically run is a design decision, not an afterthought.

01 Region

A geographic area with one or more data centres, e.g. UAE North. You choose the region when you create a resource.

02 Availability zone

Separate data centres inside one region, with their own power, cooling and network. Spread across zones to survive a data-centre failure.

03 Region pair

Many regions are paired with another in the same geography for disaster recovery and staged updates.

04 Data residency

Pick regions that meet SATORP and national requirements. Check that the AI models you need are available there.

SATORP CHECKLIST

- Which region holds the data?
- Which region runs the model?
- Are both approved for this data class?
- Is the model available in that region?
- Who signed off?

Confirm region per service before any confidential data is onboarded.

How Azure Is Organised: Four Levels

Permissions and policies set at a higher level flow down to everything inside it.

Management groups Group subscriptions to apply governance across the organisation

Subscriptions Billing and access boundary, e.g. Dev, Test, Production

Resource groups Containers for resources that share a lifecycle, e.g. one AI project

Resources The things you create: a Foundry project, a storage account, an AI Search service

PRACTICAL TIPS

- One resource group per solution and environment
- Separate subscriptions for Dev, Test and Prod
- Name resources consistently, e.g. rg-satorp-hseagent-prod
- Delete a resource group to clean up a whole lab in one step

Assign access at the highest sensible level, never wider.

Five Ways to Work with Azure

Every tool talks to the same Azure Resource Manager, so results are consistent.

01 Azure portal

Web interface at portal.azure.com. Best for learning, exploring and one-off tasks.

02 Azure CLI

Command line, e.g. az group create. Great for scripts and repeatable tasks.

03 Azure PowerShell

PowerShell cmdlets for teams already automating Windows workloads.

04 Cloud Shell

CLI and PowerShell in the browser, already signed in. Nothing to install.

05 Infrastructure as code

Bicep or Terraform files describe resources so every environment is built the same way.

06 Azure Resource Manager

The single control layer behind all of these. It checks identity and policy on every request.

Learn in the portal. Run production from code.

Microsoft Entra ID: The Front Door for Everyone

One identity service for people, applications and AI agents.

Microsoft Entra ID answers two questions on every request: who are you, and what are you allowed to do?

- **Authentication:** proving who you are, with a password plus multifactor authentication (MFA).
- **Authorisation:** deciding what you may access once you are signed in.
- **Single sign-on:** one identity for Microsoft 365, Azure, Fabric and thousands of apps.
- **Conditional Access:** rules such as “require MFA outside the corporate network”.
- **Same identity for agents:** Copilot, Copilot Studio and Foundry agents act with the signed-in user’s permissions or their own managed identity.

WHY THIS MATTERS FOR AI

- An agent should never see more than the user asking
- Permissions on SharePoint and Fabric are respected
- Every action is attributable in the audit log

Identity is the first guardrail for every agent.

Role-Based Access Control: Least Privilege by Design

Give each person or app the smallest role, at the narrowest scope, that lets them do the job.

Built-in role	What it allows	Typical holder
Owner	Everything, including granting access to others	A small platform team only
Contributor	Create and manage resources, but not grant access	Developers in a Dev subscription
Reader	View resources, change nothing	Auditors, support staff
Service-specific roles	e.g. Azure AI User, Search Index Data Reader	Apps and agents using one service

ROLE ASSIGNMENT = 3 PARTS

- **Who:** user, group or managed identity
- **What:** a role definition
- **Where:** the scope, e.g. one resource group

Assign roles to groups, not individuals.

No Passwords in Code: Managed Identities and Key Vault

The safest secret is the one nobody has to handle.

Practice	Risky way	Azure way
App signs in to a service	✗ API key pasted into a config file	✓ Managed identity: Azure gives the app its own Entra identity, no secret stored
Storing secrets you must keep	✗ Spreadsheet, email or source code	✓ Azure Key Vault: encrypted, access-controlled, audited
Rotating keys	✗ Manual, often forgotten	✓ Rotate in Key Vault; apps read the latest value
Who used the secret?	✗ Unknown	✓ Every access logged and attributable

Every Foundry and Copilot Studio solution this week should follow this pattern.

Guardrails for the Whole Estate

Governance tools stop mistakes before they happen, not after.

01 Azure Policy

Rules applied automatically, e.g. “only allow approved regions” or “storage must not be public”.

02 Tags

Labels such as owner, cost centre and environment for reporting and cost allocation.

03 Resource locks

Prevent accidental deletion or change of critical resources.

04 Microsoft Purview

Discover, classify and label data; track lineage and audit access across the estate.

05 Microsoft Defender for Cloud

Security posture score and recommendations across your subscriptions.

06 Landing zones

A ready-made, governed foundation that new projects plug into.

Good governance makes it safe to say yes to new AI ideas, faster.

Networking Basics: Keeping Traffic Private

How an AI agent can reach SAP or OpenText without touching the public internet.

Users	Sign in with Entra ID from Teams or a web app
Virtual network (VNet)	Your private network in Azure, split into subnets
Private endpoints	Give services like Storage, AI Search and Foundry a private IP inside your VNet
API Management	One secure front door that checks identity and limits every call
ExpressRoute / VPN	Private connection from Azure to on-premises or a private SAP cloud

Key terms	
VNet	Private network in Azure
Subnet	A segment of a VNet
NSG	Firewall rules for a subnet
Private endpoint	Private IP for a PaaS service
VPN gateway	Encrypted tunnel over the internet
ExpressRoute	Dedicated private circuit

Where Code Runs: Choosing a Compute Option

Pick the most managed option that still meets the need.

Service	Model	Use it when	Example in an AI solution
Virtual Machines	IaaS	You need full control of the OS or legacy software	A legacy integration server
App Service	PaaS	Web apps and APIs without managing servers	The front end and API of an agent
Azure Functions	Serverless	Small pieces of code triggered by events	Process a new document when it lands
Azure Container Apps	Serverless containers	Containerised apps that scale to zero	Hosting an MCP server (Day 2)
Azure Kubernetes Service	Managed containers	Large container platforms with many services	A platform team running many agents

Rule of thumb: Functions or Container Apps first, Virtual Machines last.

Where Data Lives in Azure

AI is only as good as the data it can reach. These are the stores you will meet.

01 Blob Storage

Cheap, massive storage for files: PDFs, images, exports. Common home for documents an agent will search.

02 Data Lake Storage Gen2

Blob Storage with folders and fine-grained security, built for analytics.

03 Azure SQL Database

Managed relational database for structured business data.

04 Azure Cosmos DB

Fast, globally distributed NoSQL database; often stores agent conversation state.

05 Azure AI Search

Indexes documents for keyword, vector and hybrid search. The engine behind RAG (Day 4).

06 Microsoft Fabric OneLake

One logical data lake for the whole organisation, in open Delta Parquet format (Day 5).

Documents feed RAG on Day 4. Tables feed Fabric data agents on Day 5.

The Azure AI Toolbox

Ready-made AI building blocks, so teams assemble rather than invent.

01 Microsoft Foundry

One place to choose models, build agents, evaluate and monitor them (Day 3).

02 Azure OpenAI in Foundry

GPT models with enterprise security, deployed in your chosen region.

03 Azure AI Search

Retrieval engine that grounds answers in your own documents.

04 Azure AI Content Safety

Filters harmful content and detects prompt-injection attacks (Prompt Shields).

05 Document Intelligence

Reads forms, invoices and permits and returns structured fields.

06 Speech and Language

Speech-to-text, translation (including Arabic) and text analysis.

Same Azure identity, networking and monitoring apply to every one of these.

Monitoring: Knowing What Your Solution Is Doing

If you cannot see it, you cannot trust it, especially with AI.

01 Azure Monitor

Collects metrics and logs from every Azure resource, with alerts when something crosses a threshold.

02 Application Insights

Tracks requests, errors and latency inside your app. For agents: every model call, tool call and token count.

03 Log Analytics

Query all logs in one place with the Kusto Query Language (KQL).

04 Service Health

Tells you when an Azure service or region has an issue affecting you.

WHAT TO WATCH FOR AN AGENT

- Error rate of tool calls
- Response time (latency)
- Token usage and cost
- User feedback, thumbs up/down
- Unanswered questions

Agree who looks at the dashboard, and how often.

Cost Management: AI Is Affordable When You Measure It

A worked example from a CodeSizzler HR agent design, sized for 100 users.

~$270
estimated Azure cost per month

$2.70
per user, per month

11,000
questions a month (100 users × 5 a day × 22 days)

~$45
of that is model tokens (small model)

Tools that keep cost predictable

- Azure Pricing Calculator before you build
- Budgets and alerts per subscription
- Tags to show cost by project and owner
- Cost Management + Billing reports

Habits that keep cost low

- Choose the smallest model that passes your tests
- Stop or delete idle lab resources
- Right-size search and hosting tiers
- Set token limits per user or app

Estimate first, set a budget, then watch real usage. Figures above are illustrative list prices.

The Azure Well-Architected Framework

Five questions to ask about any solution, including AI agents.

01 Reliability

Does it keep working, and recover quickly, when something fails?

02 Security

Is access least-privilege, traffic private and data protected?

03 Cost optimisation

Are we paying only for value, and can we see where money goes?

04 Operational excellence

Is it deployed from code, monitored and owned by a named team?

05 Performance efficiency

Does it respond fast enough and scale with demand?

PART 1 RECAP

- Cloud = on demand, pay per use
- Shared responsibility: your data, your identities
- Region choice = data residency
- Entra ID + RBAC + Key Vault
- Private networking for plant and SAP data

Next: what is an AI agent?

02

Agent Foundations

What an AI agent is, how it thinks and acts, and how we keep it safe

- LLM basics
- Anatomy of an agent
- Tools & MCP
- Multi-agent patterns
- Human in the loop

From AI to Generative AI to Large Language Models

A large language model predicts the next word, very well, based on patterns learned from huge amounts of text.

Artificial intelligence

Software that imitates human abilities: seeing, reading, deciding

Machine learning

Learning patterns from data instead of hand-written rules

Generative AI

Models that create new content: text, images, code

Large language model

A generative model for language, e.g. GPT, behind every Copilot

Prompt

The instruction or question you give the model. Clear prompts get better answers.

Token

A piece of a word. Models read and write in tokens, and you pay per token.

Context window

How much text the model can consider at once: your prompt, documents and the chat so far.

Hallucination

A confident answer that is wrong. Grounding in your own sources reduces it.

Chatbot, Assistant or Agent: What Is the Difference?

An agent does not just answer. It plans, uses tools and takes steps towards a goal.

Capability	Classic chatbot	AI agent
How it decides	✗ Follows fixed scripts and keywords	✓ A model reasons about the goal and plans the next step
Knowledge	✗ Only what was typed into it	✓ Grounded in your documents and data, with citations
Actions	✗ Answers only	✓ Calls tools: look up stock, create a work order, send an email
Memory	✗ Forgets each turn	✓ Keeps context of the conversation and, if enabled, preferences
Control	✗ Rigid	✓ Guided by instructions, guardrails and human approval

Agent = model + instructions + knowledge + tools, working towards a goal.

The Six Parts of Every Agent

Whichever platform you use, you configure the same six things.



01 Model

The language model that reasons and writes, e.g. a GPT model.

02 Instructions

The job description: role, rules, tone, what to refuse.

03 Knowledge

Approved sources: SOPs, manuals, SharePoint, data.

04 Tools

Actions it can take: APIs, connectors, MCP servers, workflows.

05 Memory

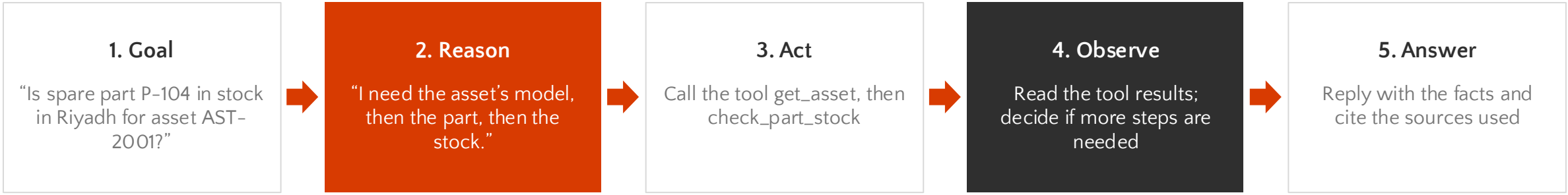
Conversation context and, optionally, saved preferences.

06 Orchestration

The planner that decides which tool or source to use next.

The Agent Loop: Reason, Act, Observe

Agents work in a loop until the goal is met. This pattern is often called ReAct.



Why it matters

- Agents can solve multi-step problems
- Every step is visible in a trace
- You can check which tool and source was used

What can go wrong

- The agent guesses instead of looking up
- It calls the wrong tool
- It loops without finishing

How we control it

- Clear instructions and tool descriptions
- Rules to stop and ask when unsure
- Test sets and traces before go-live

Grounding: Answer from Our Sources, Not from Memory

A grounded agent looks up approved content first, then writes the answer with citations.

User question	"What PPE is required for hot work in Unit 200?"
Retrieve	Search approved SOPs and procedures for the relevant passages
Augment	Add those passages to the prompt as context for the model
Generate	The model writes an answer using only that context
Cite	Show which document and section the answer came from

Good grounding habits	
Scope	Turn on "only use specified sources"
Ownership	Every source has a named owner
Freshness	Remove outdated versions
Citations	Users can click and verify
Gaps	Say "I don't know" rather than guess
Safety	HSE emergencies go to the SOP

Tools and Structured Outputs

Tools let agents act. Structured outputs let other agents and systems use the result reliably.

A tool is a function the agent may call, with a name, a description and inputs.

- **Function calling:** the model fills in the inputs; your system runs the function and returns the result.
- **Descriptions matter:** the agent chooses tools by reading their descriptions.
- **Structured output:** ask for a table or JSON with fixed fields so the next step can rely on it.
- **In Lab 1 today:** the agent writes a LOOKUP_REQUEST line and you, the human, act as the tool.

From the Lab 1 Spec Digest Agent instructions

```
Extract exactly these fields:
Feature Name, Steering Torque Range (Nm),
Operating Humidity Range (%RH), ...

Rules:
1. Only state a value explicitly written in
   the notes. Never estimate or infer.
2. If a field has two different values or is
   TBD, output 'NEEDS CLARIFICATION'.
3. If a field is missing, write 'NOT SPECIFIED'.

If you need data not in your knowledge, output:
LOOKUP_REQUEST: {"action": "...",
  "item": "...", "region": "..."}
```

Model Context Protocol: A USB-C Port for AI Tools

An open standard that lets any agent discover and call tools on another system.

Agent (MCP client)	Copilot Studio, Foundry, VS Code or another MCP-aware agent
Discover	"What tools do you have?" The server lists names, descriptions and inputs
Call	The agent calls a tool, e.g. check_part_stock(part, city)
MCP server	Wraps a real system: SAP, a work-order database, a document store

Why SATORP should care	
Build once	Wrap SAP or plant APIs one time
Reuse	Same server from Copilot Studio and Foundry
Control	Different tools per agent
Standard	No custom connector per platform
Day 2 lab	You run your own MCP server

How Much Should an Agent Do on Its Own?

Autonomy is a dial, not a switch. Set it by risk, not by excitement.

Assistive

Suggests; a person does the work.

- Drafts an email or summary
- Answers a policy question with citations
- Risk: low
- Example: shift-handover summary

Semi-autonomous

Does the steps; a person approves key decisions.

- Triages cases and drafts replies
- Stops for approval on money, safety, legal
- Risk: medium
- Example: maintenance request intake

Autonomous

Acts end to end within strict limits.

- Runs on a trigger or schedule
- Only low-risk, reversible actions
- Risk: needs strong guardrails and monitoring
- Example: weekly report pack

At SATORP, anything touching safety, money or people starts semi-autonomous, with a human approver.

Four Ways Agents Work Together

Small specialist agents beat one agent that tries to do everything.

01 Sequential chain

Agent A's output feeds Agent B. Example today: Spec Digest → Dealer Brief, chained with @mention.

02 Supervisor and workers

A supervisor splits the job, assigns it to specialist workers and combines results.

03 Hand-off

One agent passes the conversation to another that owns the topic, e.g. HR to IT.

04 Parallel review

Several agents check the same item from different angles, e.g. tone and compliance.

SEPARATION OF CONCERNS

Give each agent one job.

- Easier to test
- Easier to explain
- Fewer missed issues
- Clear ownership

One agent judging tone, law and privacy at once misses more.

Human in the Loop: Agents Recommend, People Decide

The pattern you will practise in every lab today.

01 Agent prepares	02 Human reviews	03 Human decides	04 Agent finishes
Seconds	**Minutes**	**Signed off**	**Seconds**
Extracts, classifies and drafts. Flags anything missing, conflicting or risky.	Checks the flags and the evidence. Answers lookup requests the agent cannot.	Approves, edits or rejects. Records the decision with name and date.	Applies the decision and produces the final document or reply.

Always a human decision

- Money: refunds, compensation, spend
- Safety: anything HSE-related
- Legal: threats, claims, contracts
- People: hiring, discipline, pay
- Personal data leaving its purpose

The agent does the first pass in seconds. People keep every important decision.

Writing Instructions That Work

Instructions are the agent's job description. Be specific, testable and strict about guessing.

Five parts of a strong instruction

- **Role:** who the agent is and who it serves.
- **Task:** exactly what to produce.
- **Rules:** what never to do; when to stop and ask.
- **Output format:** a table or fixed headings.
- **Boundaries:** what is someone else's job.

Example: an advisory reviewer agent

```
ROLE   You are the Compliance & Privacy Guardrail
       Agent for the SmartGuide launch.

TASK   Review drafts for personal data exposure,
       unverified claims and regional conflicts.

RULES  You only flag risk. You never approve or
       reject. If unsure whether something is
       personal data, flag it as Medium and explain.

OUTPUT A table: Excerpt | Risk Type |
        Severity (Low/Medium/High) | Recommended Fix

BOUNDARY Do not comment on tone or brand voice.
```

Responsible AI: Microsoft's Six Principles

These principles shape the guardrails built into every Microsoft AI platform.

01 Fairness

Treat people equitably; test for bias in data and outcomes.

02 Reliability and safety

Behave as intended, including in unusual situations; fail safely.

03 Privacy and security

Protect personal and confidential data; respect access rights.

04 Inclusiveness

Work for everyone, including Arabic and English speakers and different abilities.

05 Transparency

Users know they are talking to AI and can see sources.

06 Accountability

Named people own each agent and its outcomes.

In practice: content filters, prompt-injection protection, test sets, citations and a named owner.

Agent Frameworks: When You Write Code

Most of this week is low-code. Developers who need full control use these frameworks.

Framework	What it is	Typical use
Microsoft Agent Framework	Microsoft's open-source SDK for Python and .NET; succeeds Semantic Kernel and AutoGen patterns	Multi-agent workflows, MCP tools, hosted in Foundry
Semantic Kernel	Earlier Microsoft SDK for plugins, planners and function calling	Existing apps that already use it
AutoGen	Microsoft Research framework for conversations between agents	Experiments with agent teams
LangChain / LangGraph	Popular open-source libraries; LangGraph models agents as graphs	Teams with existing LangChain skills
Foundry Agent Service	Managed runtime that hosts agents and their tools in Azure	Production agents with security and monitoring

Low-code first. Reach for a framework only when a requirement forces you to (Day 3).

Four Microsoft Platforms, One Simple Rule

Start at the top. Move down only when a requirement forces you to.

If you need…	Use	Who builds it
A personal or small-team assistant over your own mail, files, Teams and SharePoint	Microsoft 365 Copilot: Agent Builder, Cowork, Notebooks	Any employee with a Copilot licence
A department or company-wide agent with connectors, workflows, approvals and analytics	Copilot Studio	Makers and IT
Answers from structured plant or business data: KPIs, tables, Power BI	Fabric data agent	Data engineers and analysts
Custom code, own model choice, private networking to SAP or OpenText, CI/CD evaluation	Microsoft Foundry	Professional developers

Each step down adds capability, and also cost, skills and operational responsibility.

03

Agentic Workflows with Microsoft 365 Copilot

Your new AI teammates, and how to build and chain your own agents

Copilot Chat | Notebooks | Cowork & Skills | Agent Builder | Agent chaining

Four Ways Copilot Works Alongside You

No coding needed. Copilot uses your identity, so it only sees what you can see.

01 **Copilot Chat in the apps**

Draft in Word, analyse in Excel, build slides in PowerPoint, summarise threads in Outlook and Teams.

02 **Notebooks**

A research binder that talks back: collect files and notes for one project and ask questions across them.

03 **Cowork**

A colleague you delegate to: give it a goal and it plans, then produces Word, PowerPoint and email outputs for you to approve.

04 **Skills**

A recipe card: a SKILL.md file that teaches Cowork your way of doing a recurring task.

WHICH TEAMMATE?

- **Ask about a set of files:** Notebooks
- **Get a multi-step job done:** Cowork
- **Same task, your way, every time:** Skills
- **Run it on a schedule:** Cowork automations

Always review what Copilot produces before you send it.

Delegating a Real Task to Cowork

Lab scenario: “Organise next Friday’s AI at Work Lunch & Learn, and make it repeatable.”

01 Notebook	02 Cowork plan	03 Approve	04 Skill + repeat
Collect	**Delegate**	**You decide**	**Reuse**
Add the event brief, speaker notes and past feedback. Ask: what is still missing?	Give the goal. Cowork proposes a plan: agenda, short deck, invite email.	Cowork pauses before sending or scheduling. You check and approve each step.	Save your way as a skill; schedule it monthly as an automation.

SATORP ideas

- Weekly production-meeting pack
- Shift-handover summary from Teams and email
- Turnaround review prep from meeting notes
- Monthly HSE bulletin draft

Notebooks gather. Cowork does. Skills make it repeatable.

Agent Builder: Five Fields Make an Agent

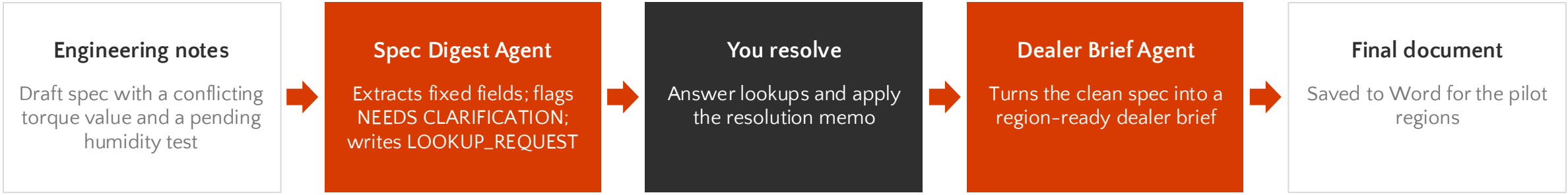
In Copilot, select Create agent, then the Configure tab.

Name	Short and specific, e.g. Spec Digest Agent
Description	One line on what it does. Copilot also uses this to pick the agent
Instructions	Role, task, rules, output format, boundaries
Knowledge	Attach files or SharePoint sites; turn on "only use specified sources"
Starter prompts	One-click prompts that show users how to start

Build → share in brief	
Build	Configure tab, then Create
Test	Try it panel; check citations
Share	Specific people or a group
Org-wide	Through the Microsoft 365 admin center
Evaluate	Manual test prompts, peer review
Grow up	Move to Copilot Studio when many users rely on it

Lab 1: Turn Messy Engineering Notes into Region-Ready Briefs

Two agents, chained with @mention in one Copilot Chat thread. About 105 minutes.



Patterns you practise

- Structured outputs
- Simulated tool calling (ReAct)
- Human-orchestrated chaining

What good looks like

- No invented numbers
- Every conflict flagged, both values quoted
- Open questions in plain language

Files you need

- Engineering notes (.docx)
- Regional requirements (.xlsx)
- Parts snapshot (.xlsx)
- Resolution memo (.docx)

Lab 2: Two Reviewers, Two Sign-offs

Agents flag risk. People approve. Each decision is documented separately.

01 Brand Voice Agent	02 Content approver	03 Compliance Agent	04 Compliance approver
Reviews clarity and tone	**Human sign-off 1**	**Reviews privacy and risk**	**Human sign-off 2**
Suggested-edits table plus an “Anything that feels out of place” list. Advisory only.	Accept or reject edits. Record: Content approved by name, team, date.	Risk table: excerpt, risk type, severity, fix. Never approves.	Decide on each risk. Record a separate, named approval.

Key lessons

- Flagging risk is not approving it
- Risk depends on where data goes, not only what it is
- Separate agents catch more
- Spot a blind spot? Extend the instructions yourself

Two specialist agents, two named humans: a governance layer anyone can build in an hour.

Challenge: Triage 12 Customer Cases Before Lunch

Northwind Home Appliances (fictional). Build, chain and supervise two agents. About 100 minutes.



Safety first

A burning smell is routed to safety immediately. The agent never troubleshoots it.

Sensitive data

A pasted card number is flagged and never repeated back to the customer.

Challenge round

Three tricky new cases test whether your instructions really hold.

What We Covered Today

Foundations first, then agents you built and chained yourself.

Azure
identity, regions, networking, cost

6 parts
model, instructions, knowledge, tools, memory, orchestration

4 platforms
M365 Copilot, Copilot Studio, Foundry, Fabric

2 + 2
agents built and humans signing off

Remember

- Your data and identities are always your responsibility
- Ground agents in approved sources; cite them
- Instructions must refuse to guess
- Agents recommend; people decide on money, safety, legal and people

Coming up on Day 2: Copilot Studio

- Build a service-desk agent with knowledge, skills and memory
- Bring your own MCP server online and connect it
- Automate fault emails with a workflow and human review
- Generate an app and publish a customer helpline

DAY 1 COMPLETE

Thank You. Questions?

Microsoft Data & AI Readiness for SATORP · Day 1: Azure & Agent Foundations

NEXT

Day 2: Copilot Studio, low-code and pro-code conversational AI. Please bring your Power Platform environment details and Copilot Studio access.

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